CO2_AT 2 – Scenario Based

Scenario 1: Manufacturing Quality Inspection

- Noise Type: Periodic Noise
- Technique: Frequency Domain Notch Filter
- Transform Used: FFT & IFFT
- Reason: Removes periodic frequencies without blurring the image.

(a) Interpret the scenario and identifying the type of noise present.

In this scenario, an automated quality inspection system captures images of metal components. Due to electrical interference, repetitive horizontal stripes appear across the images. This type of degradation is called **Periodic Noise**.

Periodic noise usually occurs because of electrical interference or electronic equipment and appears as repeated horizontal or vertical patterns in an image.

(b) Explaining how this degradation affects inspection accuracy.

Periodic noise reduces image quality by covering important details of the metal component.

Its effects include:

- Hides small cracks, scratches, and defects.
- Reduces image clarity and contrast.
- Makes automatic defect detection less accurate.
- Increases the chances of false detection or missing actual defects.

As a result, the quality inspection system becomes unreliable.

(c) Recommending an appropriate enhancement technique to remove the interference.

The most suitable enhancement technique is the **Frequency Domain Notch Filter**.

Working:

1. Convert the image from the spatial domain to the frequency domain using the **Fourier Transform (FFT)**.
2. Locate the bright frequency peaks caused by periodic noise.
3. Apply a **Notch Filter** to remove only those unwanted frequencies.
4. Convert the image back to the spatial domain using the **Inverse Fourier Transform (IFFT)**.

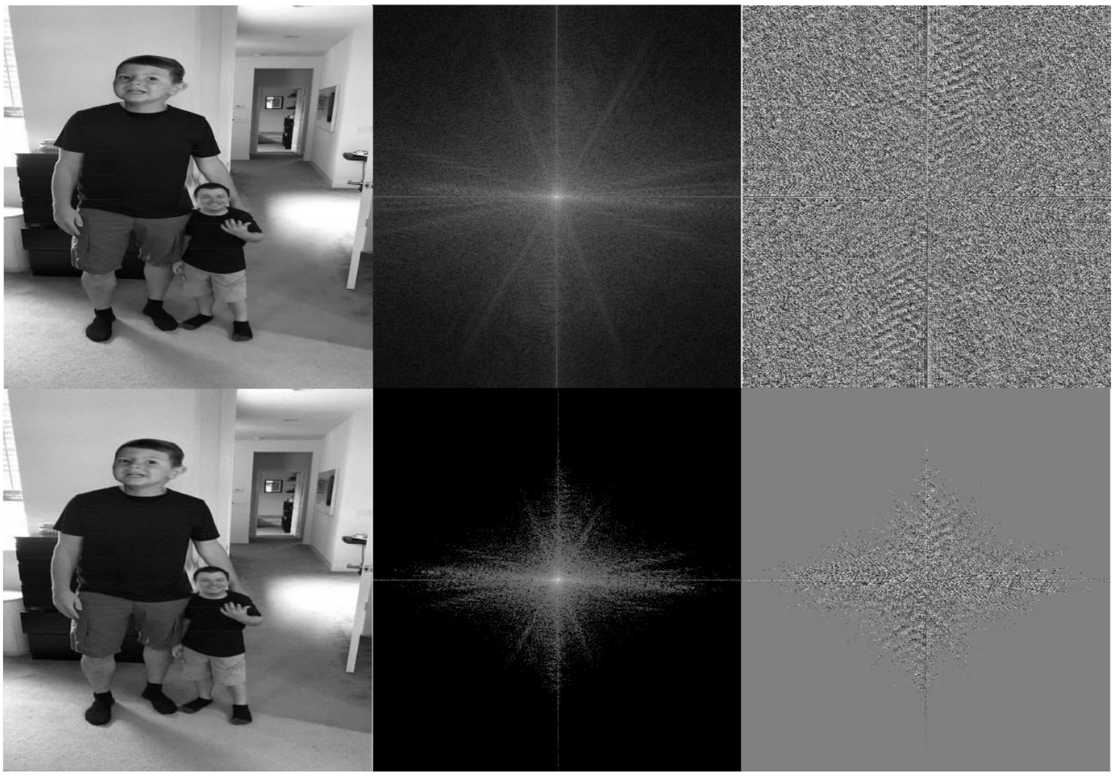
This removes the stripe noise while preserving useful image details.

(d) Justifying why our selected approach is more suitable than spatial-domain methods.

The **Notch Filter** is better than spatial-domain filters because:

- It removes only the frequencies responsible for periodic noise.
- It preserves important image features and edges.
- It avoids unnecessary blurring.
- It is highly effective for repetitive horizontal or vertical stripe patterns.

Spatial-domain filters such as mean or median filters cannot specifically remove periodic noise and may blur the entire image.



Scenario 2: Autonomous Vehicle Navigation

- Noise Type: Rain Noise
- Technique: Bilateral Filter
- Reason: Removes noise while preserving edges.
- Application: Lane detection and autonomous driving.

(a) Interpret the scenario and explain the challenge involved.

In this scenario, a self-driving car captures road images during heavy rain. Rain introduces noise into the images, making them unclear. However, lane markings and road edges must remain sharp for safe navigation.

The main challenge is to remove the rain noise without damaging important road boundaries and lane markings.

(b) Identify the importance of preserving edges while removing noise.

Edges represent important objects in the image such as:

- Lane markings
- Road boundaries
- Vehicles
- Traffic signs
- Pedestrians

If edges become blurred, the vehicle may:

- Fail to recognize lanes.
- Misidentify road boundaries.
- Make incorrect driving decisions.
- Increase the risk of accidents.

Therefore, edge preservation is essential for safe autonomous driving.

(c) Recommend a suitable filtering technique.

The most suitable filtering technique is the **Bilateral Filter**.

The Bilateral Filter smooths noisy regions while preserving edges by considering both:

- Spatial distance between pixels.
- Difference in pixel intensity.

This allows noise removal without blurring important boundaries.

(d) Justify why your selected filter preserves important image details.

The **Bilateral Filter** preserves edges because it averages only similar neighboring pixels.

Advantages:

- Removes rain noise effectively.
- Maintains sharp lane markings.
- Preserves road edges and object boundaries.
- Produces a smooth image without losing important details.

Unlike the Mean Filter, which blurs the entire image, the Bilateral Filter protects significant edges while reducing noise.